UNIT 5

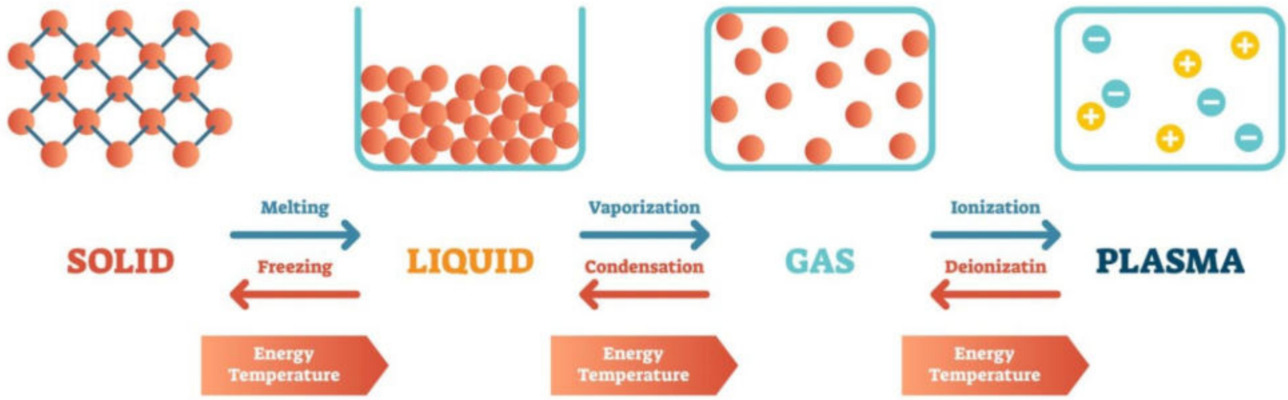
CHEMISTRY TERMINOLOGIES

Learning objectives:

- Understand liquid, gas, and plasma.
- Learn about atoms and molecule formation.
- Differentiate mixtures and balance chemical equations.

5.1 Introduction to Matter

Matter is anything that has mass and occupies space. It exists in three main states: solid, liquid, and gas. Each state has distinct properties based on how its particles are arranged.



States of Matter

• **Solid:** Their particles are tightly packed, giving a fixed shape and volume. **Example:** Ice.

• **Liquid:** Their particles are less packed, so they take the shape of its container but keep a fixed volume.

• **Example:** Water.

• **Gas:** Their Particles are spread out and move freely, with no fixed shape or volume.

Example: Air.

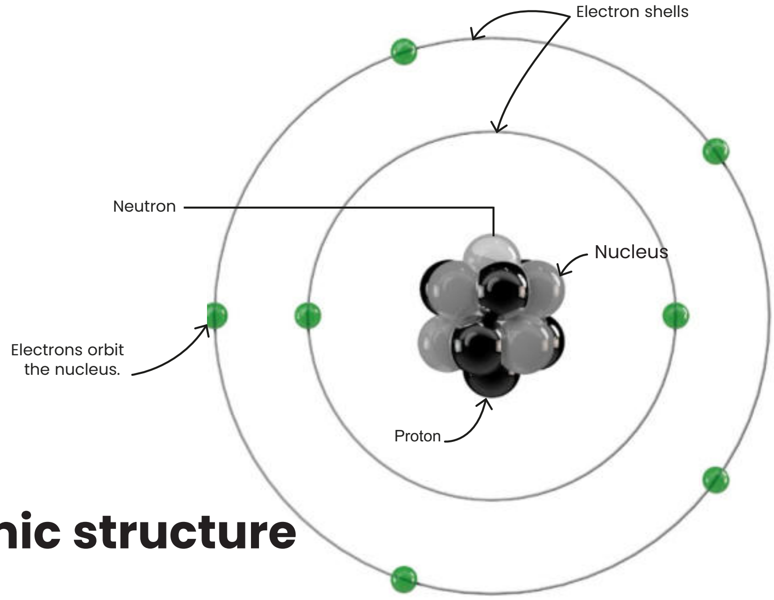
• **Plasma:** A state of matter with charged particles, found in stars and lightning.

5.2 Atoms and Molecules

Basic Structure of an Atom

An atom is made of:

- **Protons:** They possess positive charge, and are located in the central region of the atom called nucleus.
- **Neutrons:** They possess no charge, and are located in the nucleus.
- **Electrons:** They possess negative charge, and orbit the



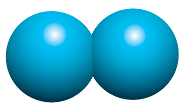
Atomic structure

Molecules: Definition and Examples

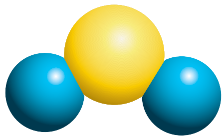
A molecule is formed when two or more atoms are combined together.

Examples:

- **O_2** Oxygen molecule (2 oxygen atoms).
- **H_2O** Water molecule (2 hydrogen atoms + 1 oxygen atom).



oxygen, O_2



Water, H_2O

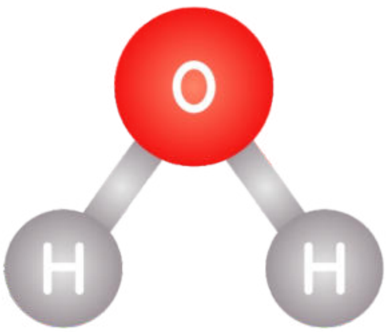
5.3 Elements and Compounds

An element is a pure substance made of only one type of atom. Each element has a unique symbol (e.g., O for Oxygen, C for Carbon and Au for gold).

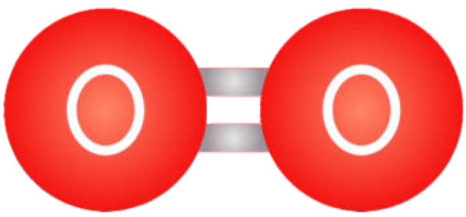
Compounds: Definition and Examples

A compound is formed when two or more elements chemically combine. Examples:

- **H_2O:** Water (Hydrogen + Oxygen).
- **CO_2:** Carbon dioxide (Carbon + Oxygen).



molecules of oxygen



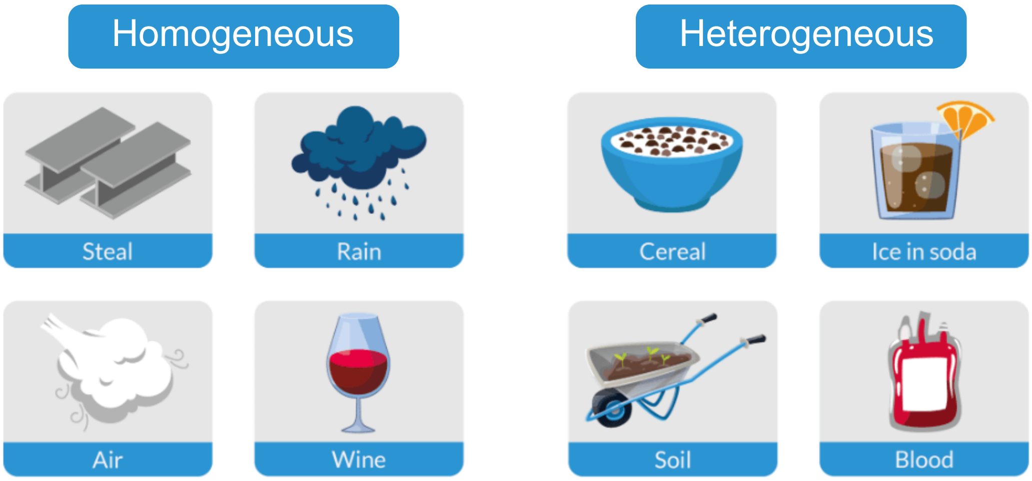
5.4 Mixtures

Homogeneous Mixtures

A mixture with a uniform composition throughout, like salt dissolved in water.

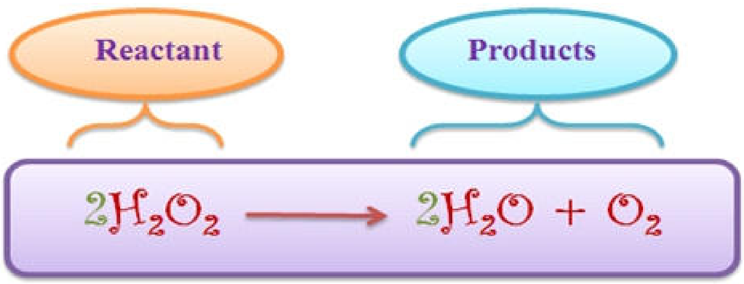
Heterogeneous Mixtures

A mixture where the components are not evenly distributed, like sand in water.



5.5 Chemical Equation Basics

A chemical equation shows how reactants transform into products.



For example:

$$2H_2 + O_2 \rightarrow 2H_2O$$

This equation represents the reaction of hydrogen and oxygen to form water.

How to Balance a Chemical Equation

An unbalanced equation

This equation shows the elements hydrogen and oxygen reacting to make water. It is an unbalanced equation, because it has two oxygen atoms on the left but only one on the right.

Key Facts

- ✓ Equations must be balanced, with the same number of atoms on both sides, because atoms are never lost or gained during reactions.
- ✓ Unbalanced equations can be balanced by adding numbers in front of the formulas, until both sides have the same number of atoms.

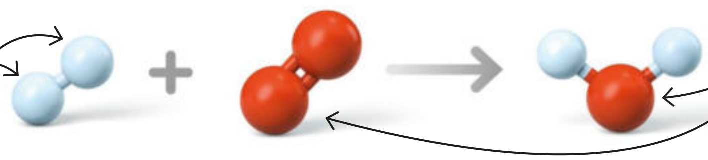
This "H" is the formula for hydrogen.

This small "2" represents two hydrogen atoms.

This "O" is the formula for oxygen.

This is the formula for water.

$$H_2 + O_2 \longrightarrow H_2O$$



How to balance an equation

Keep adding numbers to the equation until it is balanced.

1. Add a big "2" to the left-hand side next to hydrogen. This doubles the number of hydrogen atoms.

2. Add a big "2" to the right-hand side. This adds another water molecule. Recount. This now matches the number of atoms on the left-hand side.

$$2H_2 + O_2 \longrightarrow 2H_2O$$

A balanced equation ▲

It is a balanced equation, because it has two oxygen atoms on the left and two on the right. And four hydrogen atom on both sides.

EXERCISES:

Fill in the blanks.

1. Matter has ____________ and occupies ____________.

2. A ____________ is formed by atoms bonding together.

3. In solids, particles are ____________ packed.

4. The smallest unit of an element is an ____________.

5. A mixture in which components are unevenly distributed is a ____________ mixture.

6. Oxygen and nitrogen are examples of ____________.

7. Water's formula is ____________.

8. In a chemical equation, substances on the left are ____________.

Tick(✓) for True and cross (x) for False statements.

1. Matter has mass and occupies space. ☐

2. Water is an element. ☐

3. A compound forms when elements combine. ☐

4. A homogeneous mixture has unevenly distributed components. ☐

5. Protons have a negative charge. ☐

6. Chemical equations only show products. ☐

7. Gases have fixed shape and volume. ☐

Choose the correct answer.

1. What is the smallest unit of matter?

a) Proton   b) Atom

c) Electron   d) Molecule

2.Which is a homogeneous mixture?

a) Oil and water   b) Salt in water

c) Sand in water   d) Rocks and soil

3. Which is a compound?

a) O_2 b) H_2

c) N_2 d) CO

4. Which is a chemical equation?

a) $2H_2 + O_2 \rightarrow 2H_2O$ b) $H_2O + O \rightarrow H_2O_2$

c) $H_2O + CO_2 \rightarrow O_2$ d) Both a and b

5.Plasma is a form of:

a) Solid   b) Liquid

c) Gas   d) Charged gas

6. Which state of matter has particles that are spread out and move freely?

a) Solid   b) Liquid

c) Gas   d) Plasma

7. What type of mixture is formed when salt dissolves in water?

a) Homogeneous b) Heterogeneous

c) Colloid d) Suspension

8. Which of the following represents a chemical reaction?

a) Melting of ice b) Burning of wood

c) Freezing of water d) Dissolving sugar in tea

Answer the Following Questions:

Short-Answer Questions.

1. What is matter?

2. How many states of matter are there?

3. What is basic structure of an atom?

4. Write the properties of protons, neutrons, and electrons.

5. Write the properties of states of matter.

6. Define molecules, elements and compounds.

7. Write the differences between homogeneous and heterogeneous mixture.

Creating a Model of an Atom

ACTIVITY

Create a model of an atom using balls to represent protons, neutrons, and electrons. Label each part and explain the structure to the class.

Critical Thinking

Why is it important to understand the states of matter when studying chemical reactions? How do these states affect reactions in daily life?

Work sheet

Balancing Chemical Equations

Balance the equations below by adding the correct coefficient in front of each compound.

1. ___ $H_2O \rightarrow$ ___ H_2 + ___ O_2
2. ___ $Fe_2O_3 \rightarrow$ ___ Fe + ___ O_2
3. ___ Na + ___ $Cl_2 \rightarrow$ ___ $NaCl$
4. ___ $HgO \rightarrow$ ___ Hg + ___ O_2
5. ___ $TiCl_4$ + ___ $O_2 \rightarrow$ ___ TiO_2 + ___ Cl_2
6. ___ Al + ___ $O_2 \rightarrow$ ___ Al_2O_3
7. ___ HCl + ___ $Zn \rightarrow$ ___ H_2 + ___ $ZnCl_2$
8. ___ Al + ___ $CuCl_2 \rightarrow$ ___ $AlCl_3$ + ___ Cu

Homogeneous or Heterogeneous Mixture or Pure Substance

Identify each material as either a mixture or pure substance.

• If the material is a mixture, identify it as either homogeneous or heterogeneous.

• If the material is a pure substance, identify it as an element or a compound.

Material	Mixture or Pure Substance	Homogeneous or Heterogeneous / Element or Compound
1. Aluminum foil		
2. Air		
3. Soil		
4. Water (H_2O)		
5. Steel		
6. Bag of M&M's		
7. Sugar		
8. Sugar water		
9. Pizza		
10. Blood		

Material	Mixture or Pure Substance	Homogeneous or Heterogeneous / Element or Compound
11. Table Salt (NaCl)		
12. Iron Filings		
13. Gasoline		
14. Coffee		
15. Orange Juice		
16. Pencil lead		
17. Copper		
18. Bronze		
19. Milk and Cereal		
20. Acetic acid		